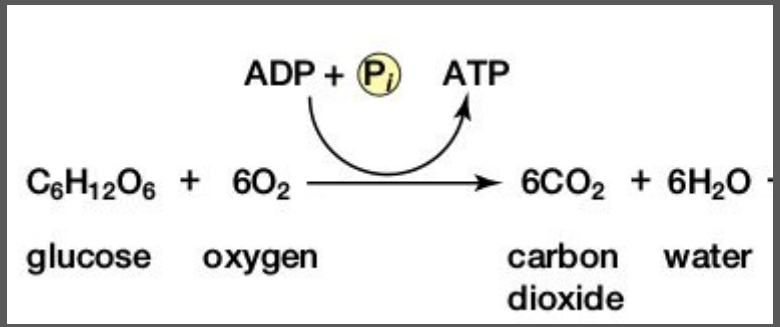


Cellular Respiration

Cellular Respiration Overview

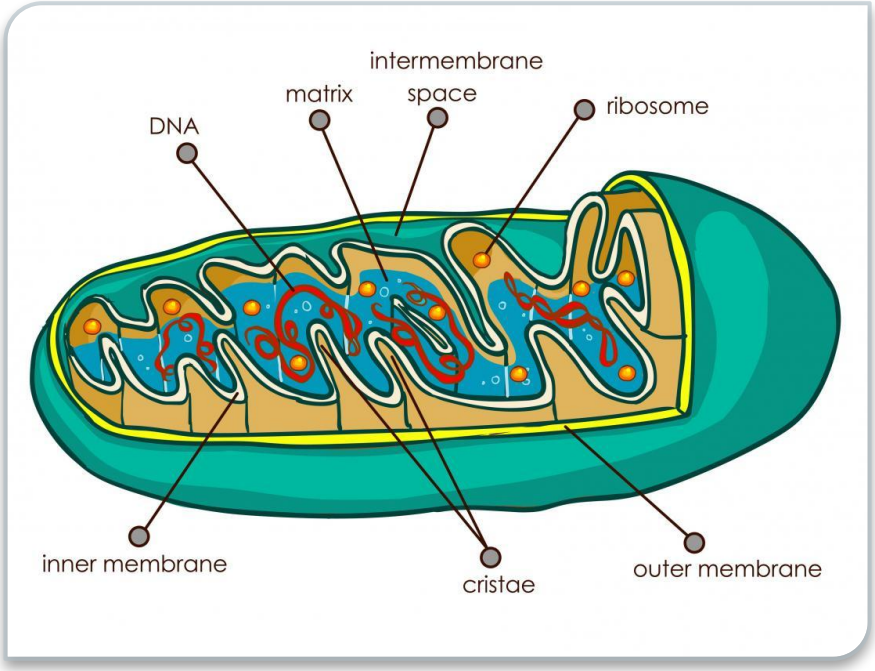
- **Goal**: to make energy in the form of ATP
  - Energy in carbs (glucose) are converted and stored in ATP
- Carbohydrates are used first as a source of energy, but any food can be broken down as a source of energy
- **Reactants**: Glucose and Oxygen
- **Products**: Carbon Dioxide and Water



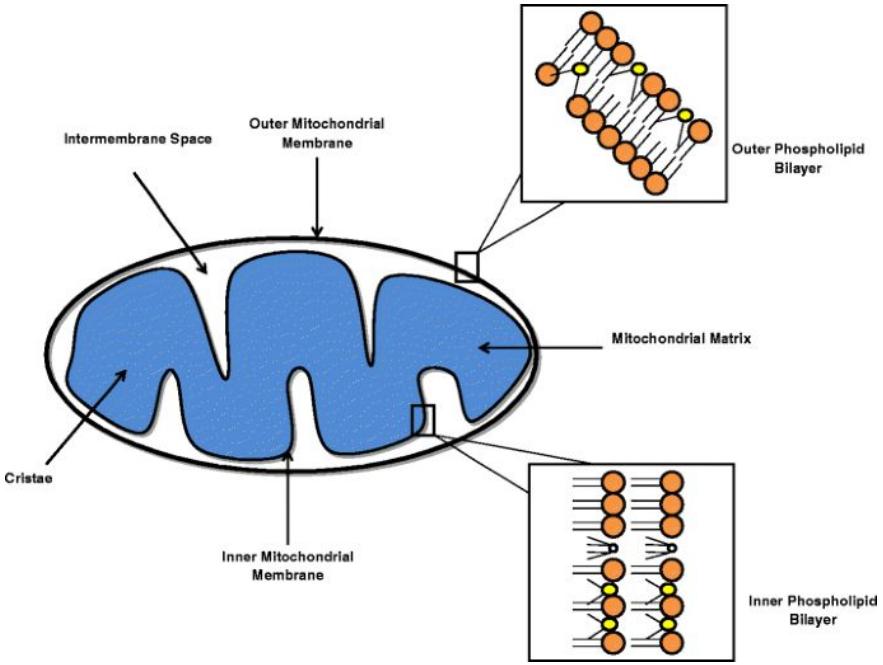
Note: Energy is released in the form of ATP but is usually **NOT** considered a product!

Mitochondria Structures

- Double membranes: outer and inner
- **Inner membrane**: folded membranes inside of the mitochondria
  - **Cristae**- inner membrane folds that increase surface area to increase ATP production
- **Intermembrane space:** fluid-filled space between membranes where H+ gradient builds
- **Matrix**: fluid-like substance that fills the space
- Contain their own DNA and Ribosomes!



Lipid Bilayers



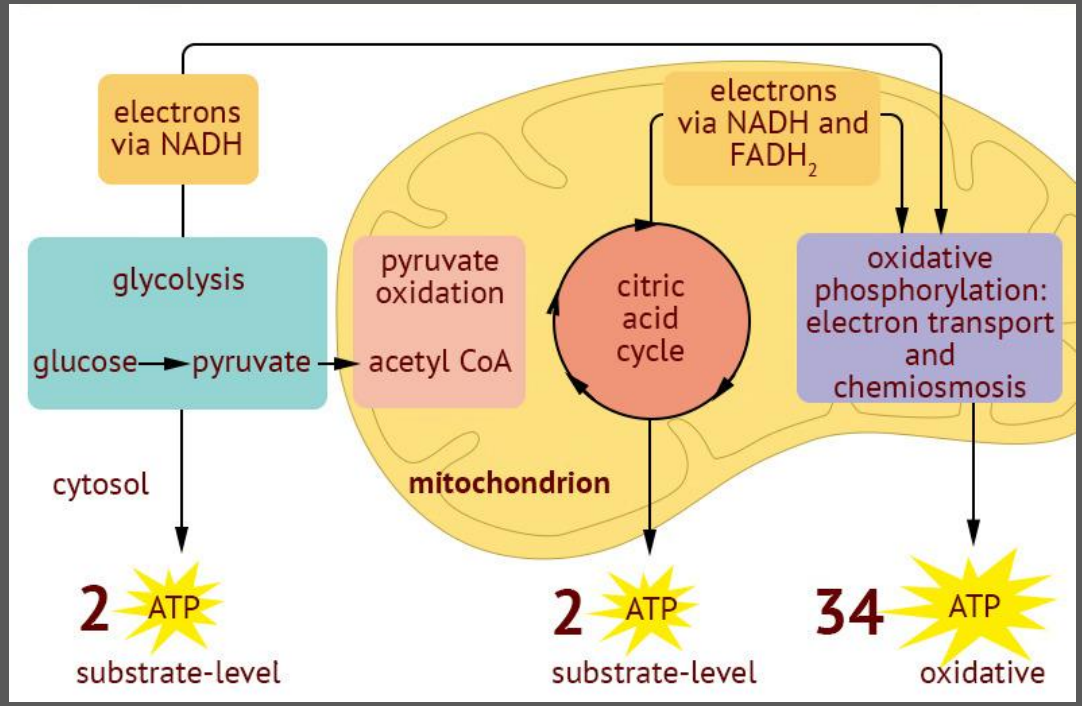
Cellular Respiration Overview

- 4 Metabolic Stages
- **An**aerobic Respiration
  - Respiration without O2
  - Occurs in Cytosol
- Aerobic respiration
  - Respiration with O2
  - Occurs in mitochondria

1. Glycolysis

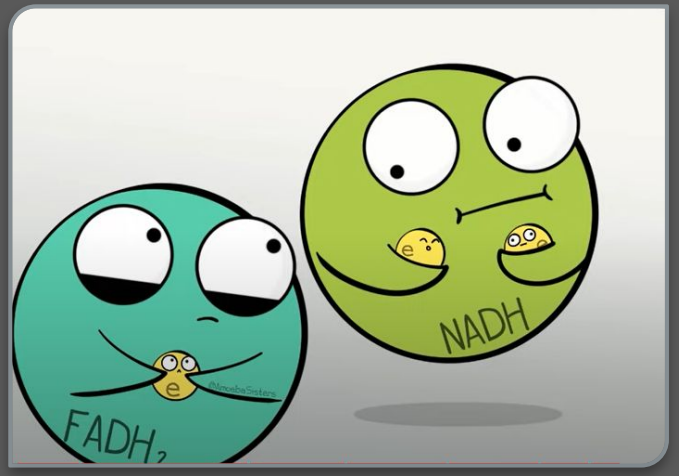
2. Pyruvate Oxidation
3. Krebs Cycle
4. Electron Transport Chain

Overview of Cellular Respiration

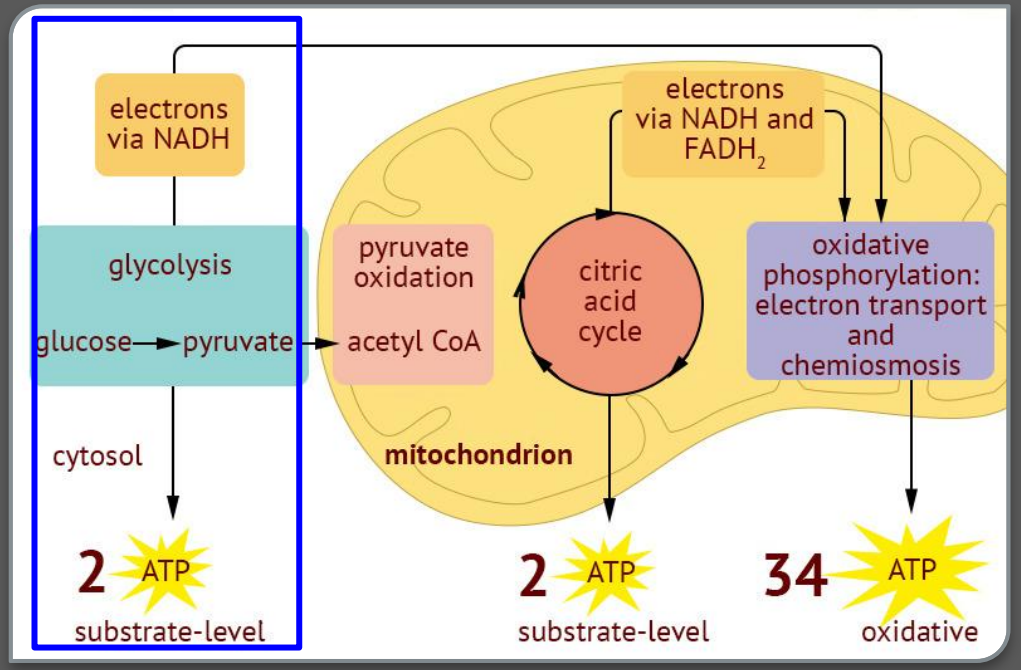


Cellular Respiration Overview

- Gradual **oxidation** of glucose releases the energy to make ATP at each step
- Electrons and hydrogens are carried down to the last step of aerobic respiration **(Electron Transport Chain)**
- **NAD+ and FAD** are coenzymes that act as electron carriers for the cell and shuttle electrons to the ETC.
    - NAD+ is **reduced** into NADH and FAD is **reduced** into $FADH_2$

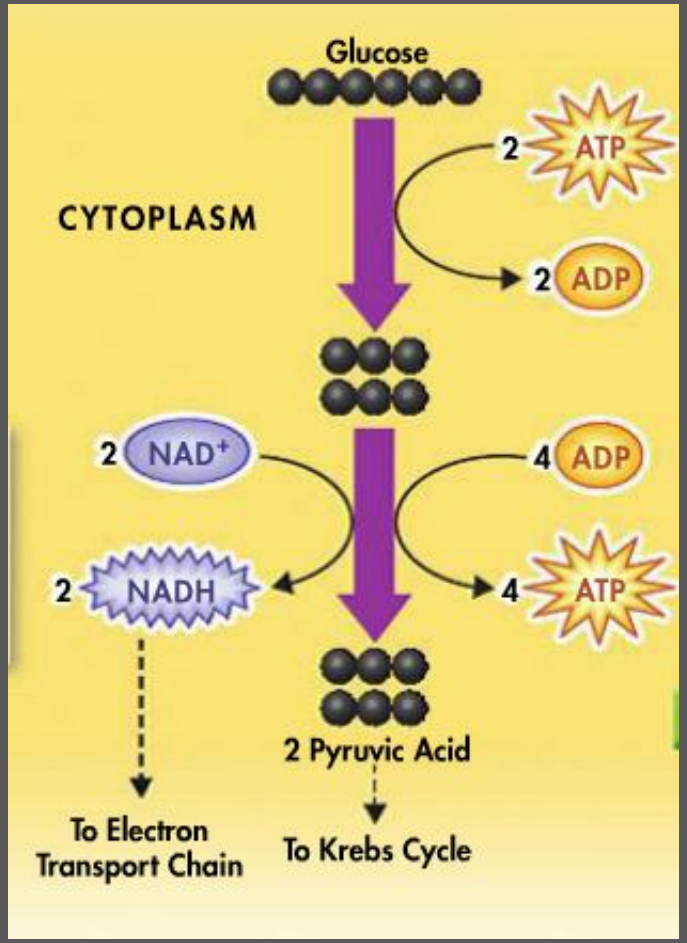


Step 1: Glycolysis



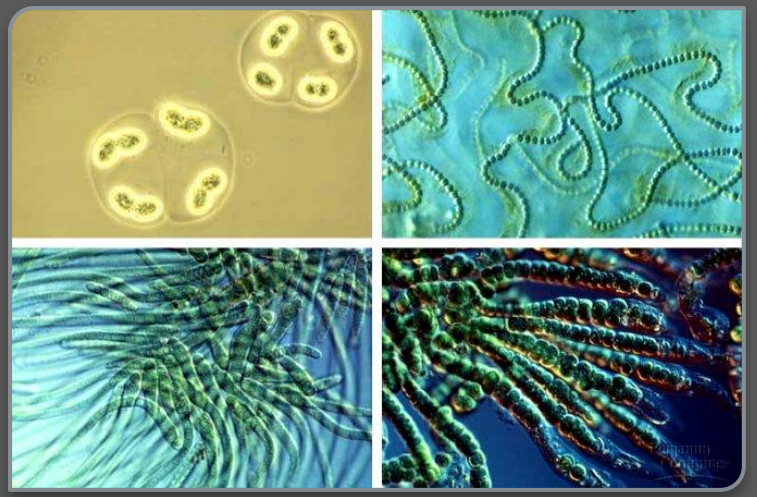
Glycolysis Overview

- Glycolysis: breaking down glucose into two molecules of pyruvate
- Occurs in cytosol
- **An**aerobic- occurs without oxygen
- Inefficient: generates only 2 ATP per 1 glucose
- 10 step process catalyzed by enzymes



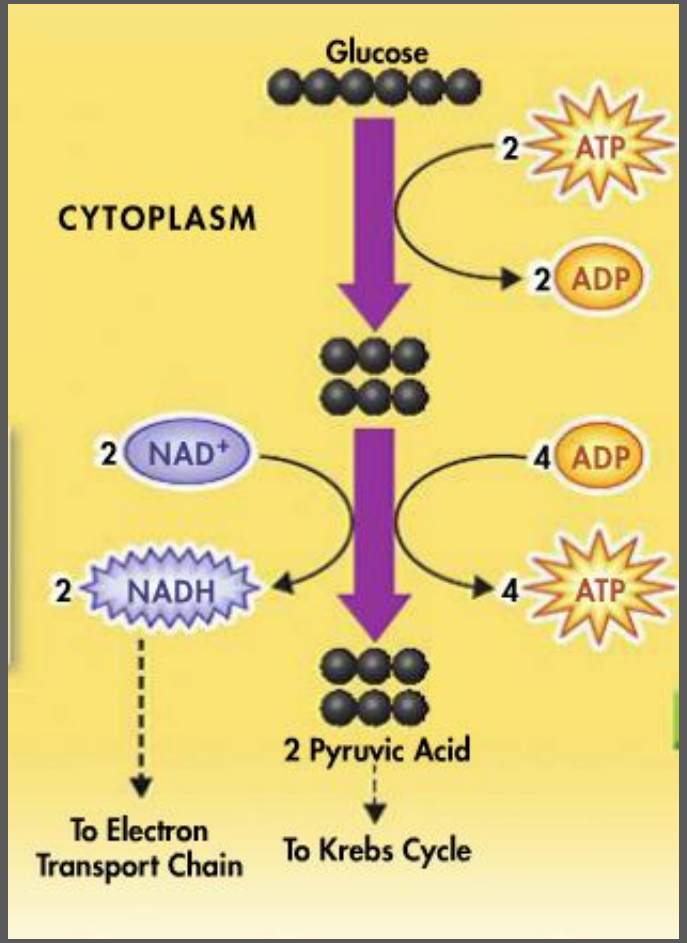
Evolutionary Perspective- Why is it Anaerobic?

- Prokaryotes were the first cells (our ancestors) and had no organelles
- Life first evolved **without** free Oxygen in the atmosphere
- Prokaryotes evolved glycolysis without mitochondria
- **All cells use glycolysis!!**

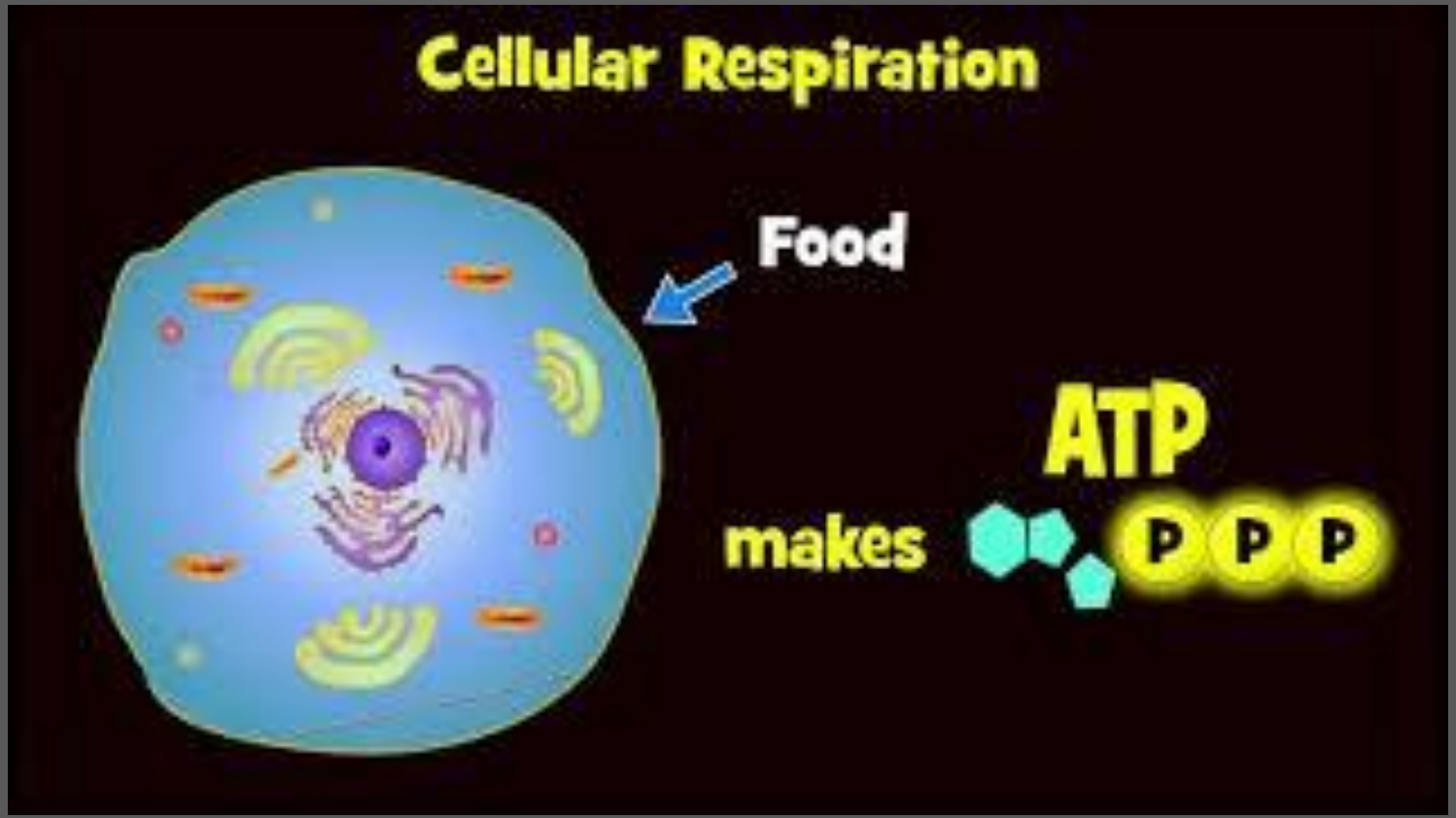


Steps to Glycolysis

1. Two phosphate groups get added to glucose (a 6 carbon molecule) from 2 ATP molecules
2. The phosphate groups make glucose unstable and it **breaks** into two 3-carbon molecules (G3P)
3. This 3 carbon molecule gets **oxidized** to make 2 NADH and 2 pyruvate are created. (4 ATP total get made in this process)

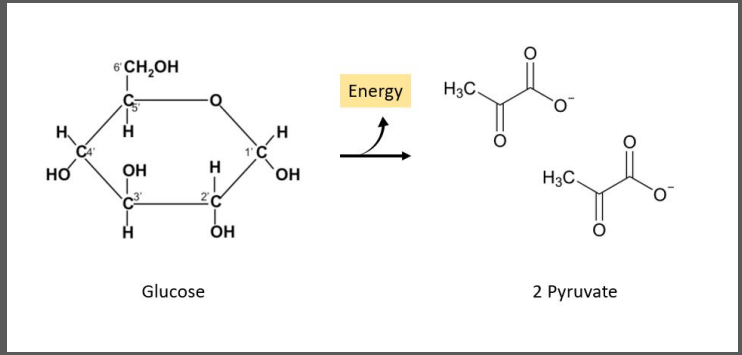


Glycolysis video- watch me!!

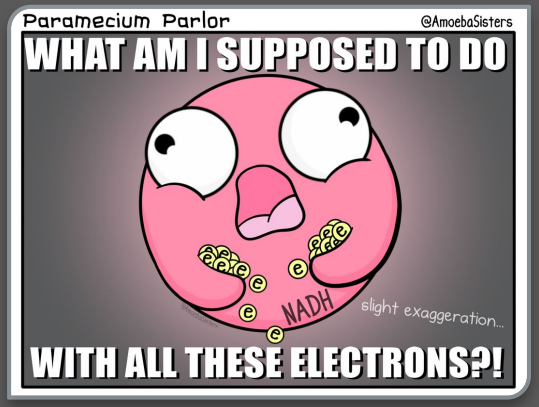


Glycolysis Summary

- Converts: 1 glucose to 2 pyruvate
- Produces: 4 ATP & 2 NADH
- Consumes: 2 ATP
- Net Yield: 2 ATP and 2 NADH



Where does the NADH go? → carries electrons and H+ down to the ETC!



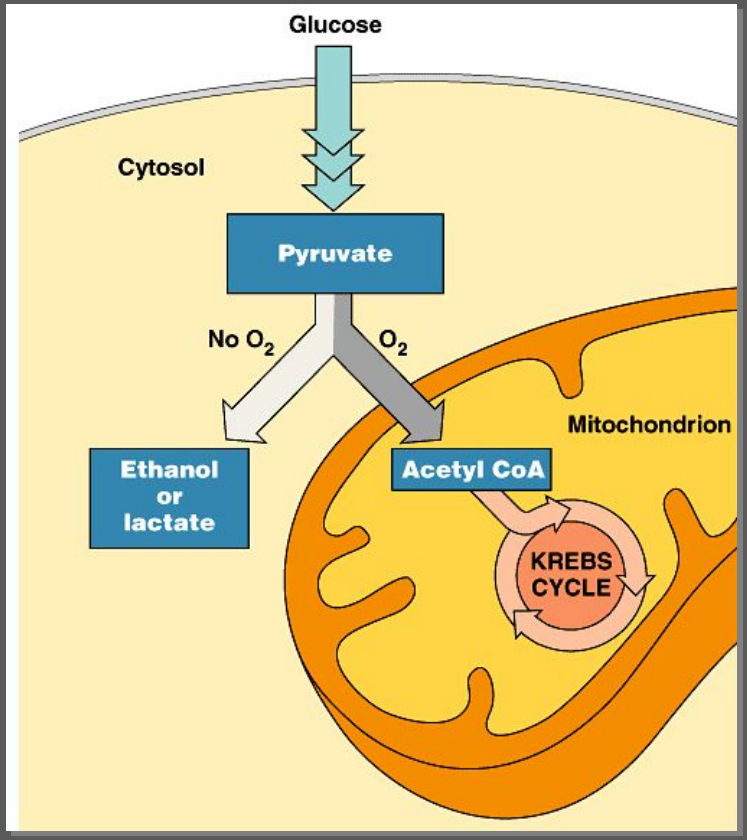
Branching Point-Decision Time!

If oxygen is present:

- aerobic respiration takes place in the mitochondria

If NO oxygen present:

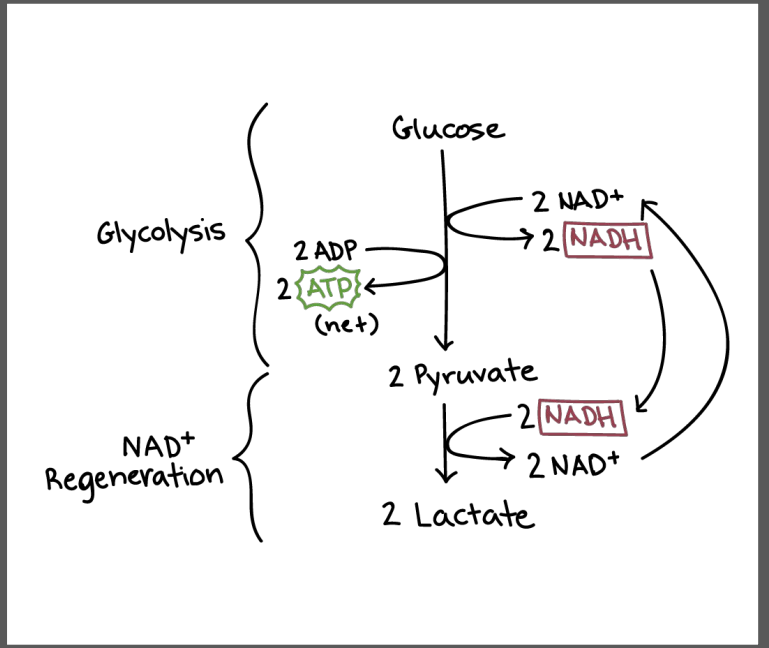
- anaerobic respiration (**fermentation**) takes place in the cytoplasm
- 2 Types: Alcohol and Lactic Acid Fermentation





Lactic Acid Fermentation

- Performed by animals and some fungi
- Pyruvate is converted to lactate and 2 ATP
- Significance: NADH is converted back to NAD+ to be recycled back into glycolysis (**NAD+ Regeneration**)
- Reversible once O2 is present again, lactate is converted back to pyruvate by the liver



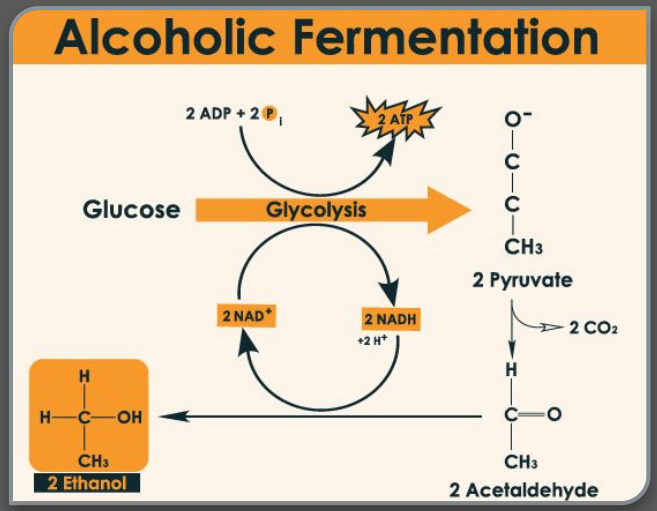
In humans, this occurs when your muscles cannot get enough oxygen to keep up with demand

Alcohol Fermentation

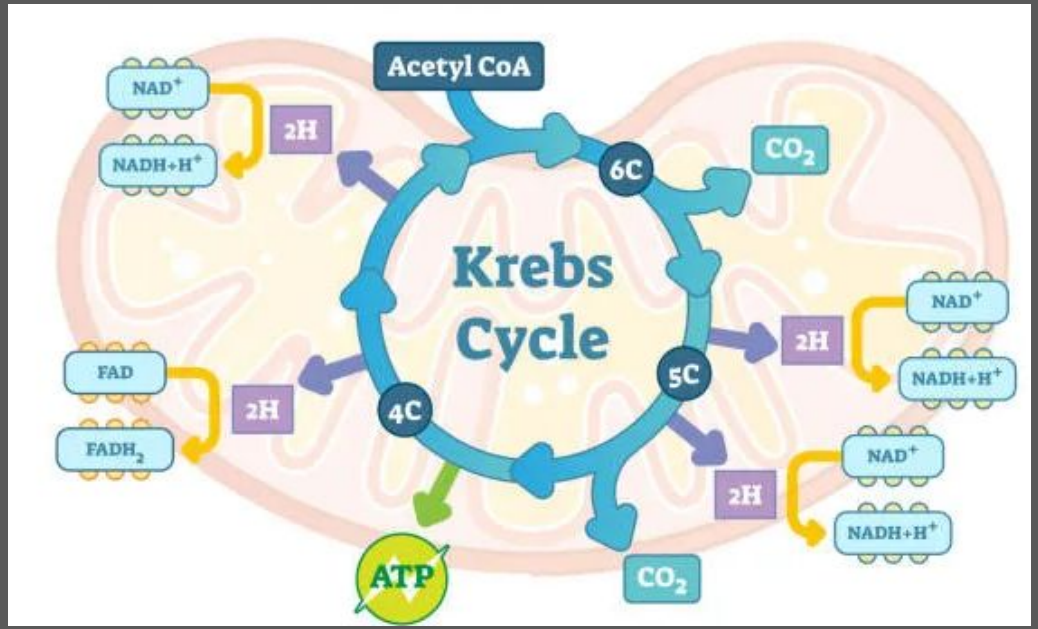




- Occurs in a single celled eukaryote called yeast, and in bacteria
- Used commercially for baking and alcohol production
- Pyruvate is converted to **ethanol and CO2**
  - NADH is converted back into NAD+ to be used again in glycolysis
- Produces only 2 ATP
- Dead end process: at ~12% ethanol yeast cells die and you cannot reverse this reaction!

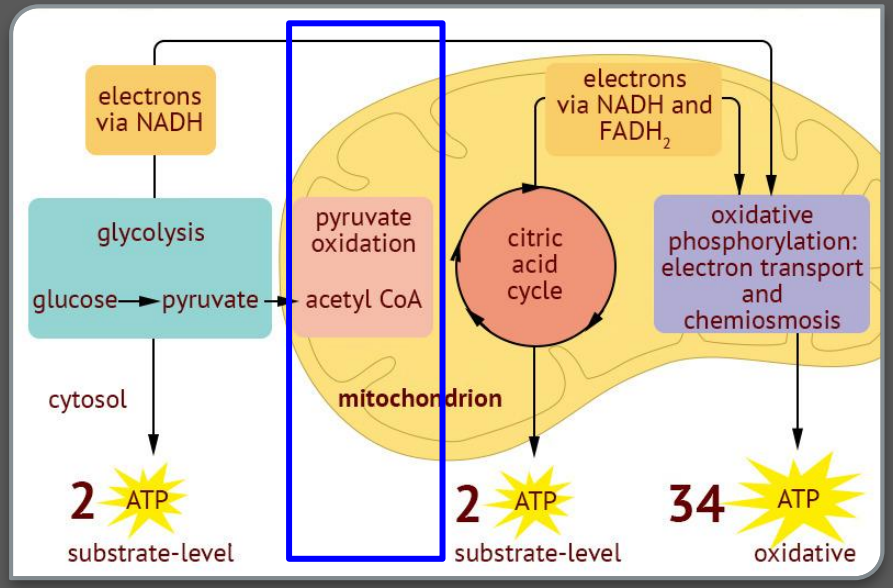


Steps 2-3: Pyruvate Oxidation and the Krebs Cycle



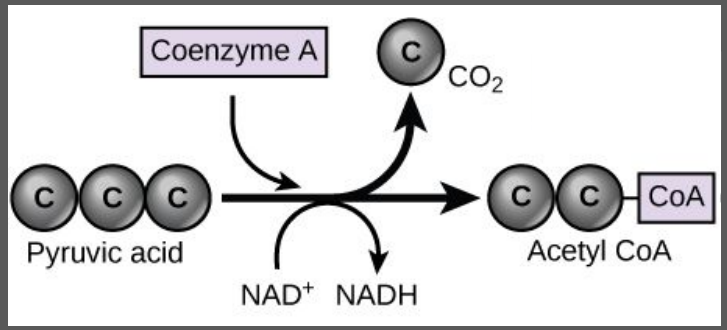
Step 2: Oxidation of Pyruvate

- Pyruvate still has energy to yield
- If O_2 is available, pyruvate enters the mitochondria specifically the mitochondrial matrix



Oxidation of Pyruvate Steps

- Pyruvate is converted into Acetyl Coenzyme A and CO2 (2x!)
  1. Pyruvate loses a carbon and 2 oxygens released as CO2 **(decarboxylation)**
  2. NAD+ is reduced to NADH
  3. Coenzyme A is transferred to pyruvate **helps bring Acetyl to Krebs Cycle



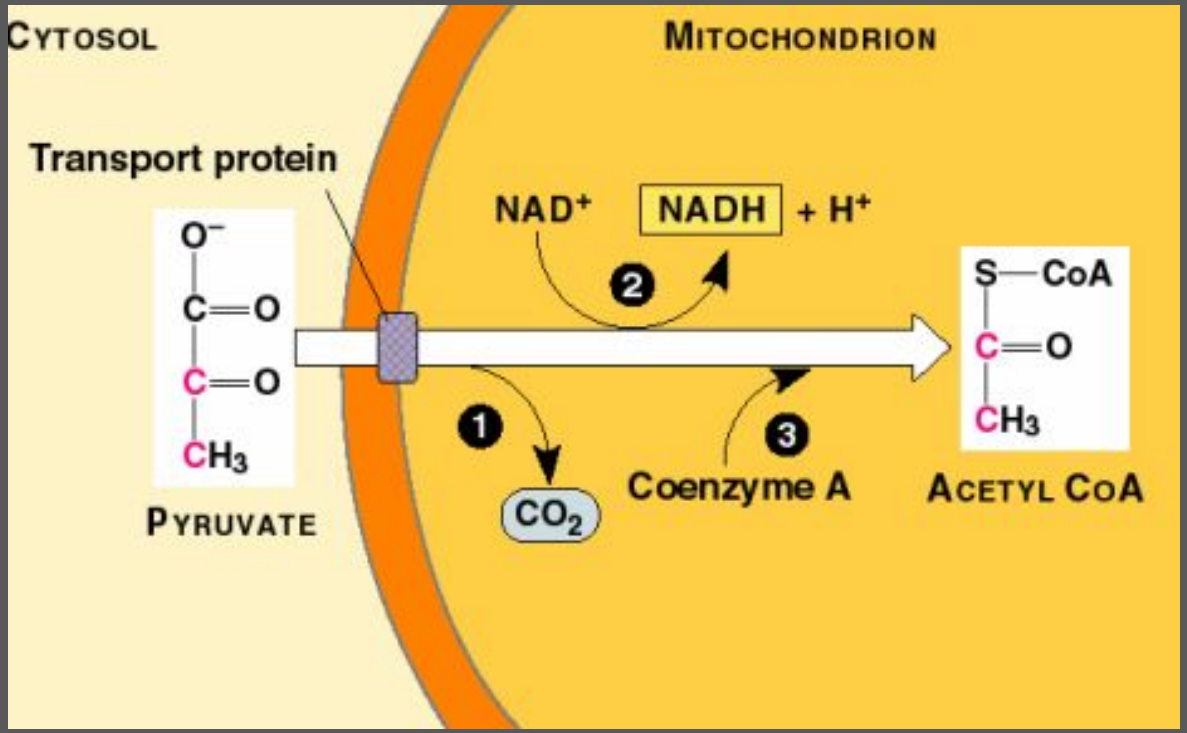
- 2 acetyl CoA is produced and moves on to **Krebs cycle**
- NADH gets saved for the **ETC**

Pyruvate Oxidation Video



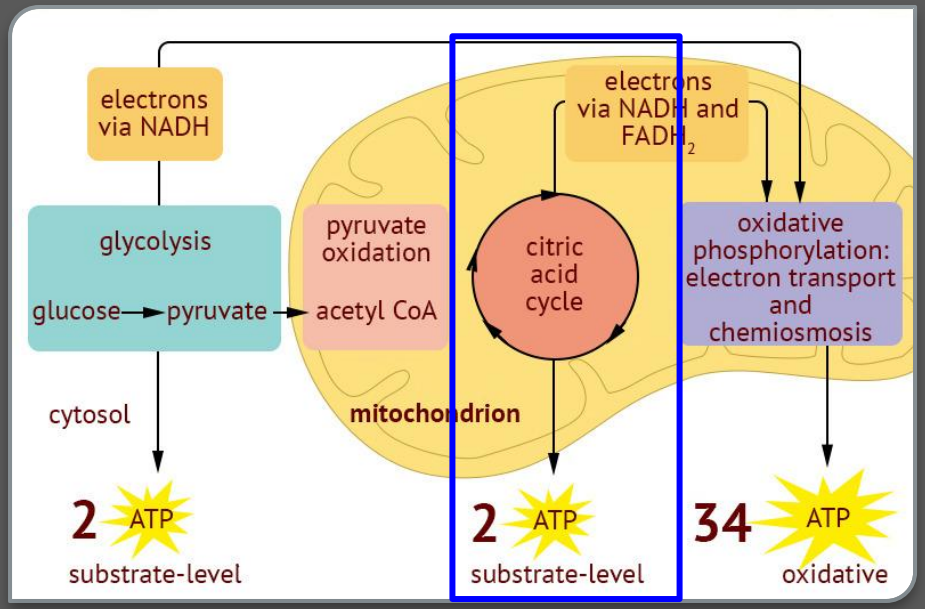
Pyruvate Oxidation in Detail

Where does CO_2 go?



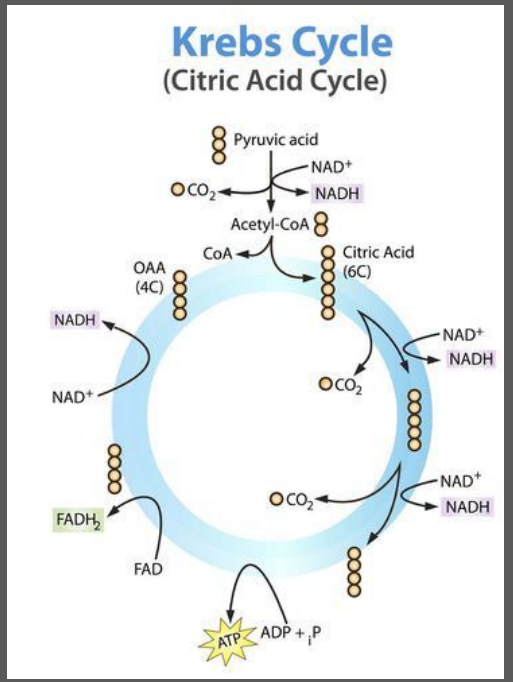
Step 3: Krebs Cycle

- AKA Citric Acid Cycle
- Identified by Hans Krebs in 1937
- Happens 2x per 1 glucose
- Goal: Make a large amount of electron carriers for ETC
- Acetyl-CoA is oxidized repeatedly in order to reduce electron carriers (NAD+ and FAD)

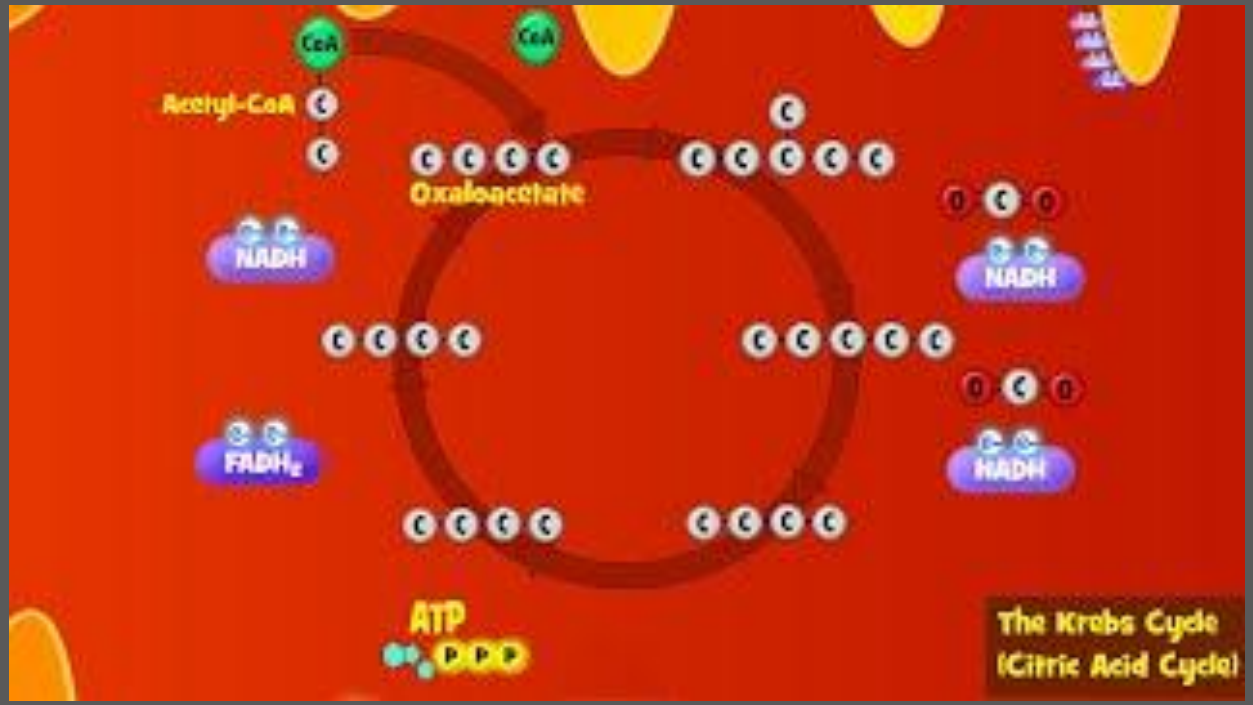


Stages of the Kreb Cycle

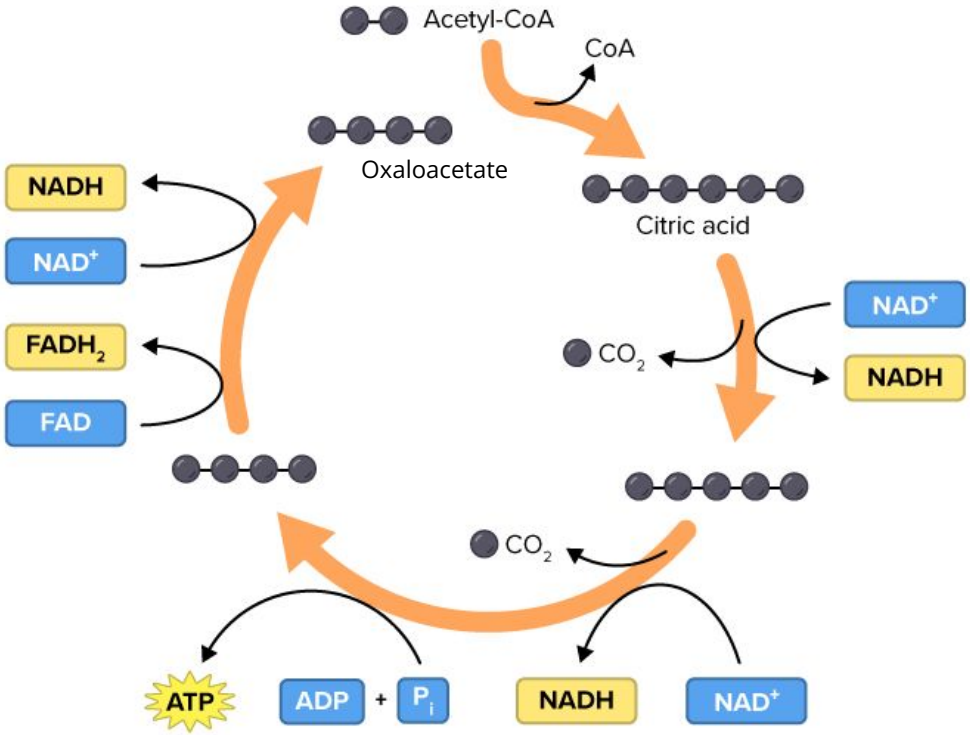
1. Acetyl-CoA is combined with Oxaloacetate to form Citric Acid and CoA is removed
2. Citric acid is oxidized many times and electrons are transferred to electron carriers (NAD+ and FAD)
    - Carbons are also removed in this process and CO_2 is released
3. Molecules are rearranged to form oxaloacetate again to be <u>continuously</u> used in the Krebs Cycle



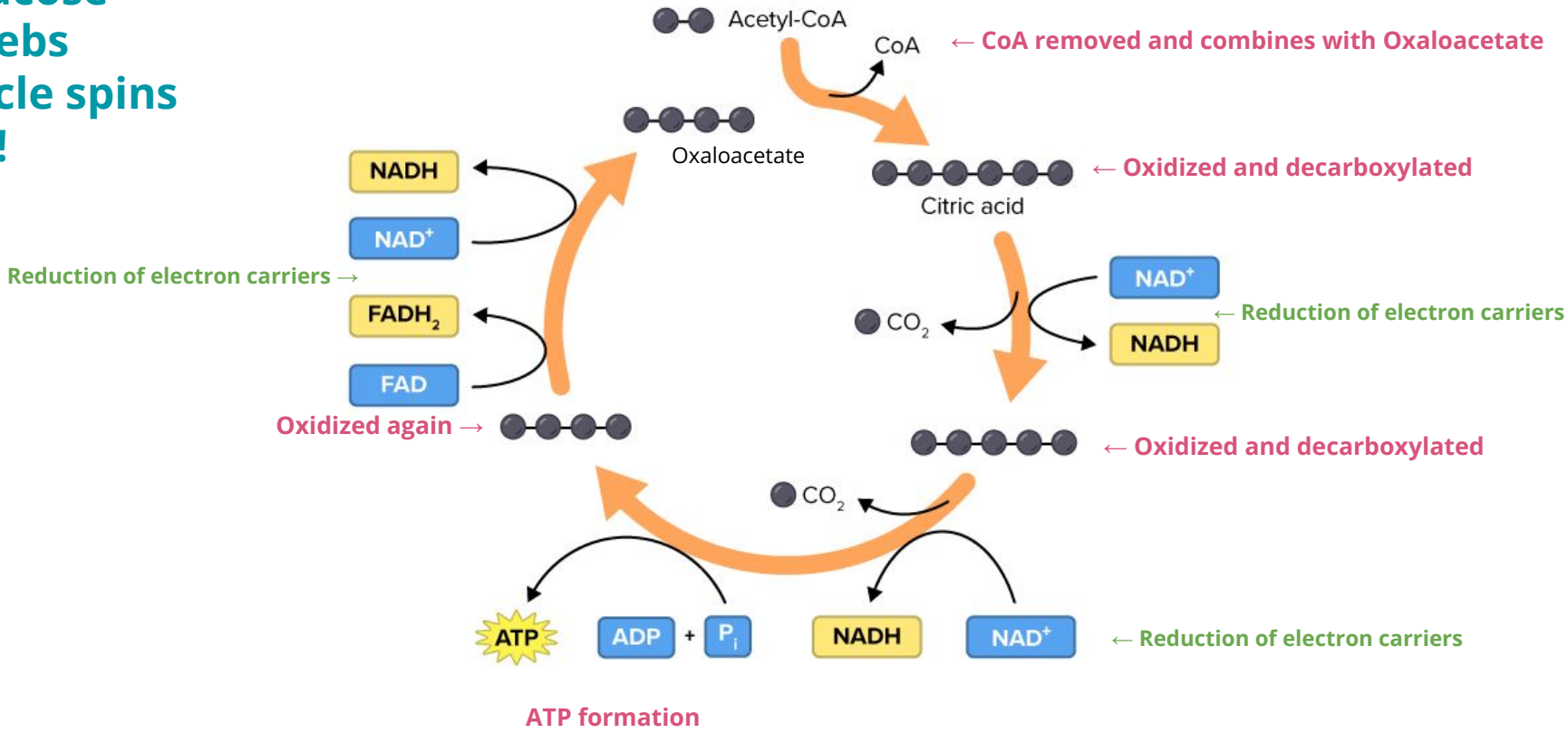
Krebs Cycle Video



Per 1 glucose Krebs cycle spins 2x!



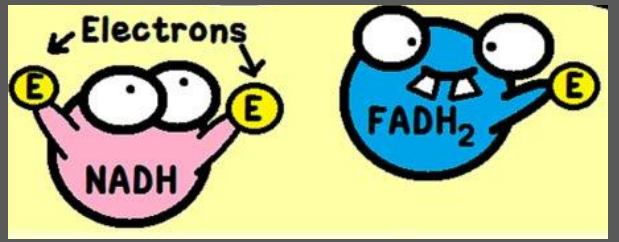
Per 1 glucose Krebs cycle spins 2x!



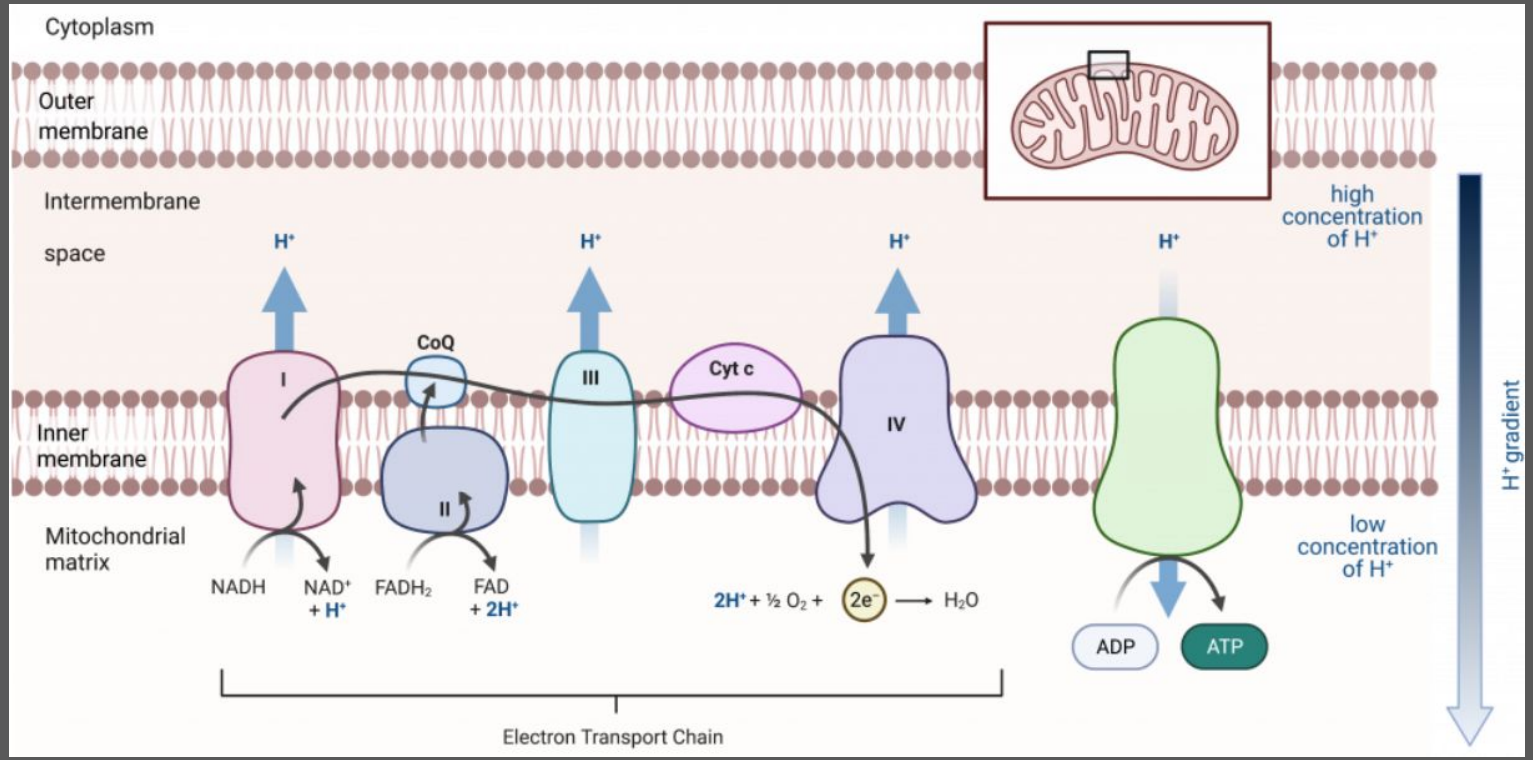
Summary of the Krebs Cycle

- Spins **2x** per 1 glucose
- Input: 2 Acetyl CoA
- Outputs:
  - 2 ATP
  - 6 NADH
  - 2 $FADH_2$

Where will NADH and $FADH_2$ go? To the ETC to be used to make A LOT of ATP



Stage 4: The ETC

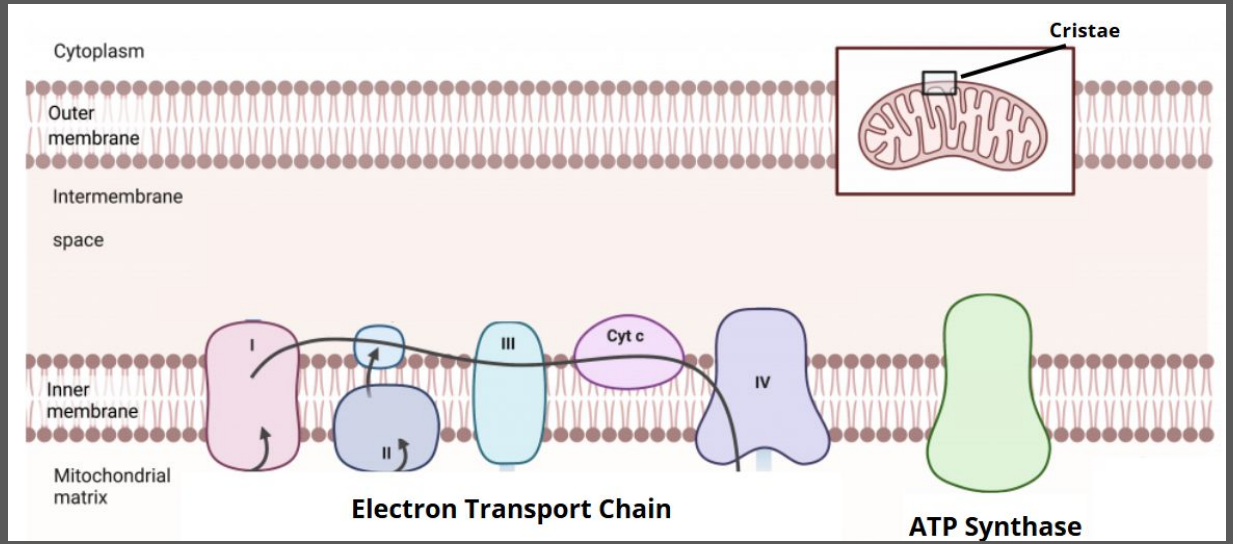


Summary of Reactions So Far

- **Outputs of Steps 1-3**
  - **ONLY 4 ATP** → we need MUCH more to support multicellular organisms
  - 10 NADH
  - 2 $FADH_2$
- We now have many stored electrons and H+ ions (protons) that we will use to power the last step!

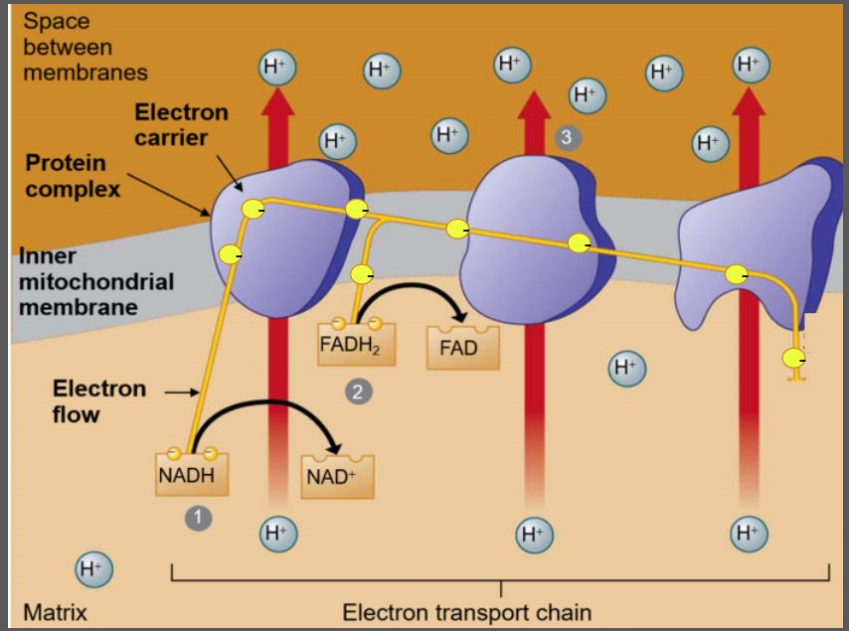
Structures of the ETC

- Location of the ETC: **Cristae** (folds of inner membrane)
- Electron transport chain: series of protein complexes that move electrons and pump H+ ions from the matrix to the intermembrane space
- ATP Synthase: an enzyme that generates ATP



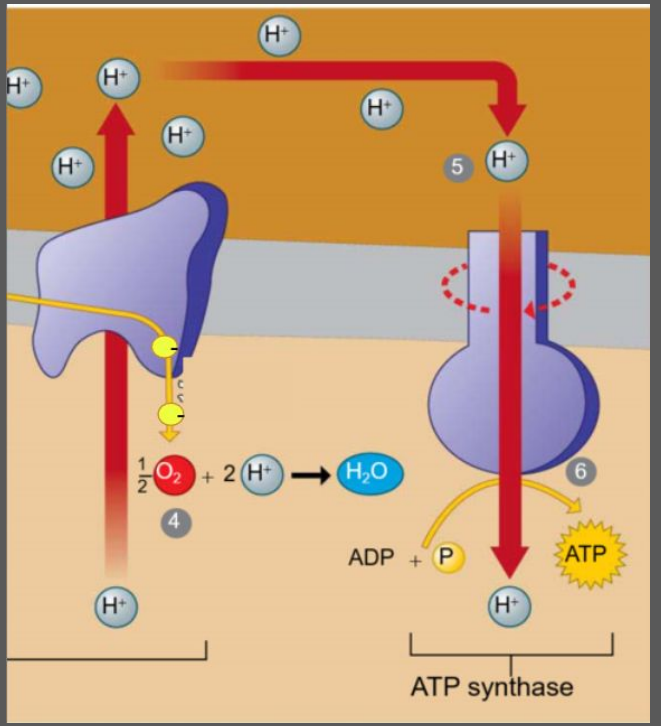
Stage 1: Building a Proton Gradient

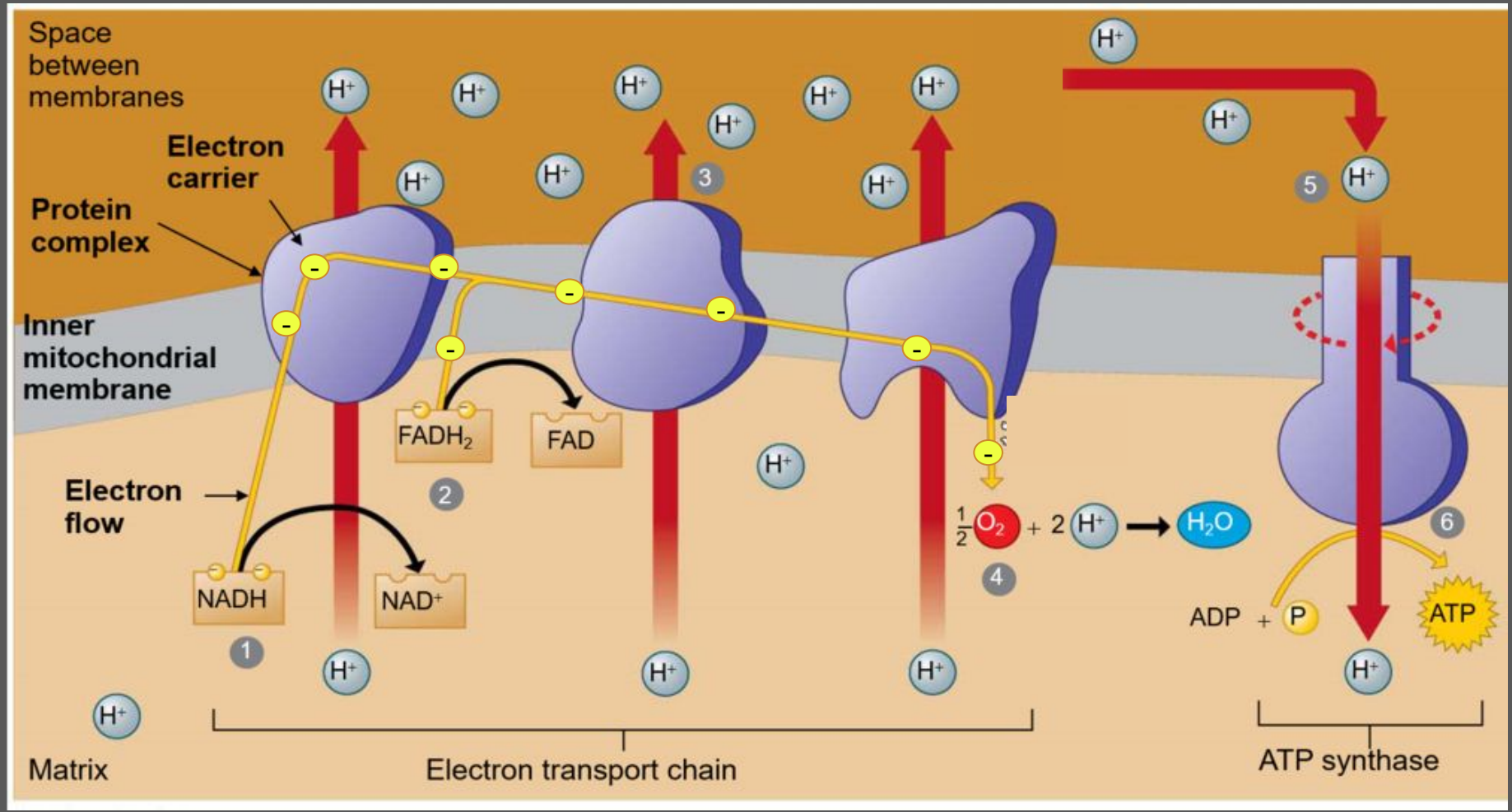
1. **H atoms** are removed from the electron carriers NADH and FADH2 **(oxidation)**
    - NAD+ and FAD can be recycled into previous steps!
2. Electrons delivered to the first protein in the ETC
3. Electrons move from one protein to the next along the inner membrane and energy is released. These proteins pump H+ (protons) from matrix into intermembrane space to create a H+ gradient



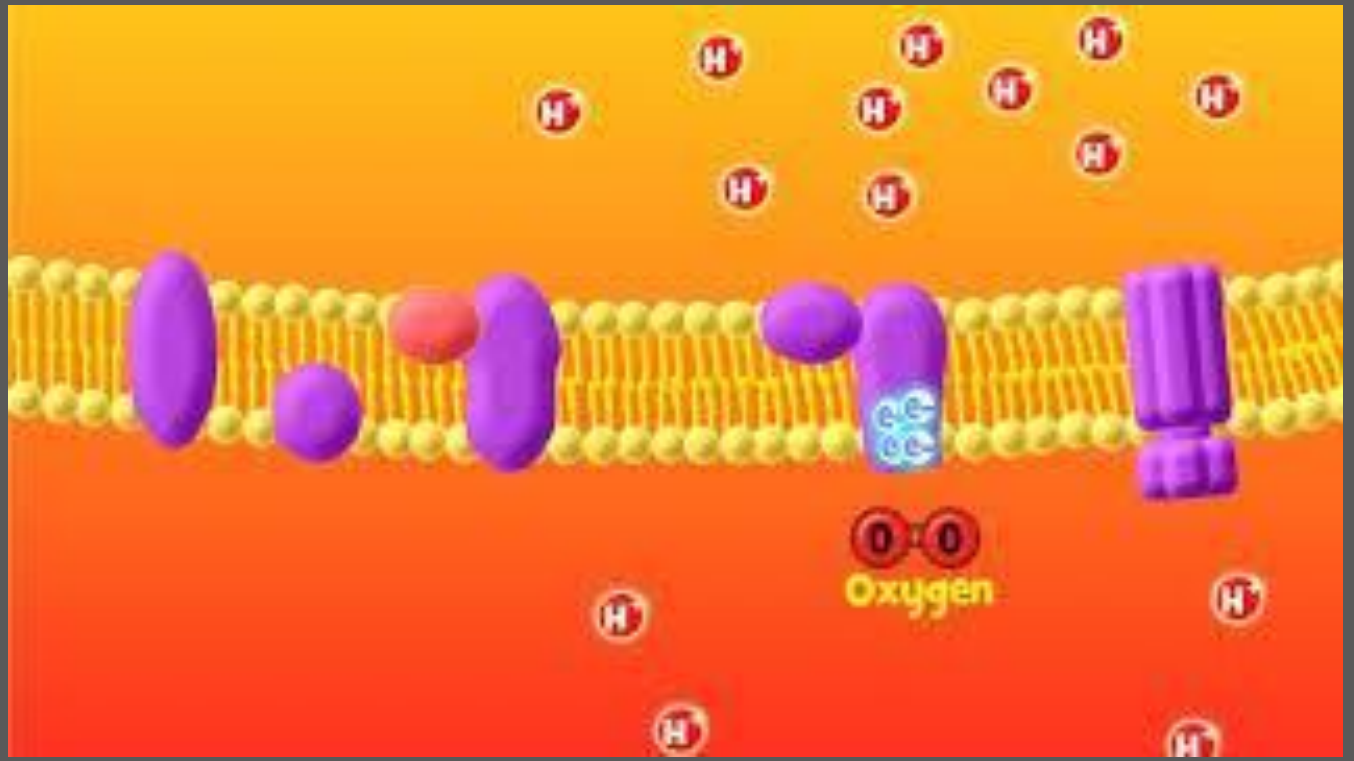
Stage 2: Oxidative Phosphorylation

1. H+ diffuses back into the matrix through ATP synthase which spins like a turbine to phosphorylate ADP to ATP
2. H+ and e- are reunited and combine with O_2 to make H2O (O_2 is the “final electron acceptor”)
    - Electrons (e-) are “pulled” down the ETC because O2 is highly electronegative and strongly attracts electrons!




Space between membranes
Electron carrier
Protein complex
Inner mitochondrial membrane
Electron flow
NADH
NAD+
FADH2
FAD
H+
1
2
3
4
5
6
½ O2 + 2 H+
H2O
ADP + P
ATP
Matrix
Electron transport chain
ATP synthase

ETC Video



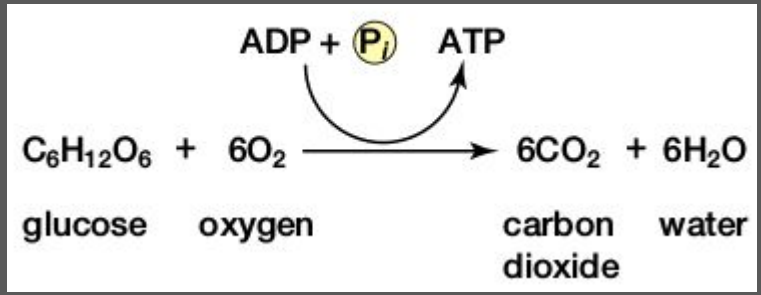
What Happens Without Oxygen?

- Electrons have nowhere to go → oxygen isn't there to accept them.
- The ETC backs up with electrons.
- NADH and $FADH_2$ can't drop off their electrons
  - The cell runs out of NAD^+ and FAD, which are needed for glycolysis and the Krebs cycle to continue.
- As a result, ATP production from the ETC stops.

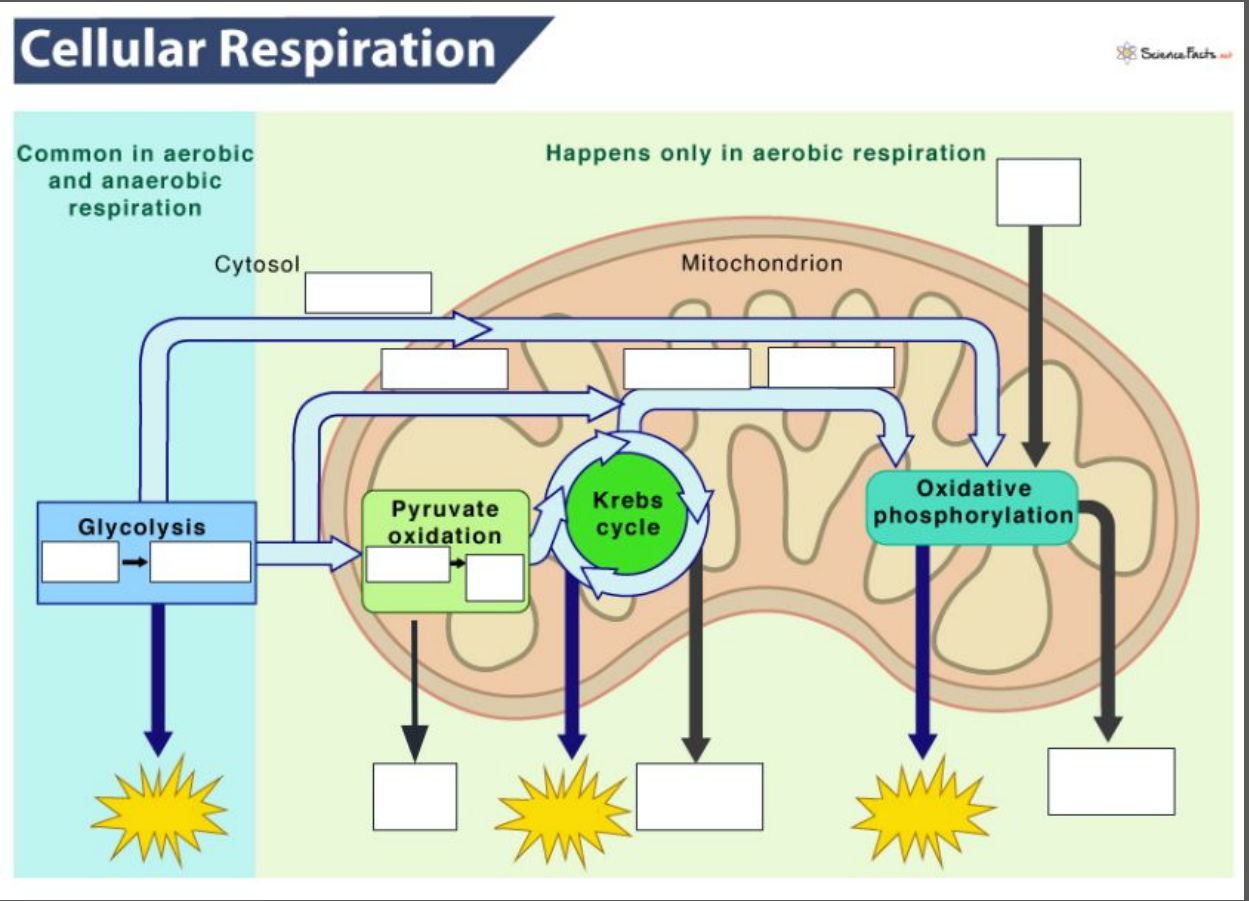
What process does the cell revert to as a back to in the absence of oxygen? What is the goal of this process?

Summary of Cell Respiration

- Where did the glucose come from?
- Where did the O2 come from?
- Where did the CO2 come from?
- Where did the CO2 go?
- Where did the H2O come from?
- Where did the ATP come from?
- What else is produced that is not listed
- in this equation?
- Why do we breathe?



Let’s summarize the outputs below!



Let's summarize the outputs below!

